Group #69

Traceability matrix

ID	Requirement	Related Use Case	Fulfilled By	Description	Implemented By	Tested By
1	AED should power on and perform self-test.	UC1: Power ON	AEDSimulator	Ensures the AED is operational and ready for use by conducting a self-test that checks device functionality and preparedness when powered on.	AEDSimulator	Run the program simulation in Qt and press the power on button on the display
2	AED should analyze heart rhythm.	UC3: Cardiac Arrhythmia Detection	AEDSimulator	AED must analyze the heart rhythm to detect irregularity in the heartbeat and decide on the need for a shock, dependent on if the rhythm is shockable or not (sinus or asystole)	AEDSimulator, Patient	After the electrode pads are connected, the heartbeat is checked and the display informs the user of the cardiac arrhythmia
3	AED should provide real-time CPR feedback.	UC4: Real-Time CPR Feedback	AEDSimulator, Patient	AED provides feedback on CPR efforts, guiding the user to perform effective resuscitation and offering feedback if the resuscitation attempt is lackluster.	AEDSimulator	While CPR is being administered by the User to the patient, the display provides feedback as is needed for the user to either do more compressions or if they are doing good compressions
4	AED should advise when a shock is necessary.	UC3: Cardiac Arrhythmia Detection	AEDSimulator, User	AED must analyze the heart rhythm and advise the user whether a shock is necessary or whether to administer CPR immediately.	AEDSimulator	There is a button present that will light up to represent that a shock is necessary after a shockable rhythm has been detected

5	AED should allow the user to deliver a shock.	UC5: Shock Delivery	AEDSimulator	AED allows for the delivery of a shock when it is deemed necessary, ensuring the user can respond promptly.	AEDSimulator	After the shock button is lit up to show a shock is necessary, the user is prompted to press it for the shock to be delivered to the patient`
6	AED should provide post-shock care instructions.	UC7: CPR and Post-Shock Care	AEDSimulator	AED provides instructions for post-shock care, guiding the user on subsequent steps after a shock has been given.	AEDSimulator	
7	AED should track and update the patient's heart beat.	UC3: Cardiac Arrhythmia Detection	AEDSimulator	AED maintains an up-to-date record of the patient's heart rhythm for accurate analysis and response.	AEDSimulator, Patient	After every shock and administration of CPR, the heart rhythm is then analyzed again to determine whether an additional shock is needed or CPR or if the rhythm has stabilized
8	AED should be able to receive and process CPR input.	UC4: Real-Time CPR Feedback	AEDSimulator, User	AED receives and processes user input during CPR to provide appropriate feedback for the user to perform more effective CPR.	User, AEDSimulator	While CPR is being administered by the User to the patient, there is a slider present to represent CPR compressions and the display provides feedback as is needed for the user
9	AED should provide a user interface for interaction.	All Use Cases	MainWindow	AED features a user interface that facilitates interaction with all system functions and user inputs.	MainWindow	Once the program simulation is running in Qt, the entire interface is displayed with fully functional buttons accounting for every possible heart rhythm and directs the user to applying the solution using the interface

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